

CLPV : Dataset of Controlled Co-Located Lighting and Photometric Variation for Robot Vision

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Abstract—Robust robot vision in low-light environments depends not only on scene conditions, but also on image-acquisition choices. Camera exposure, gain, and onboard illumination directly shape the measurements available to downstream perception, yet robotics datasets seldom vary these sensing parameters in a controlled, systematic way. We present CLPV , a dataset and capture platform for studying controlled photometric variation in robot vision. The fixed-view component comprises 136 static scenes, each recorded across five exposure times, four gain settings, and eleven co-located illumination levels. This yields 220 hardware-synchronized stereo pairs of identical scene geometry, accompanied by ambient and per-intensity lux measurements. The mobile component includes 18 repeatable robot routes and 13,667 hardware-synchronized stereo pairs with LiDAR, odometry, and full-route lux traces. An autonomous capture workflow enables repeated traversal of each route under fixed-exposure and auto-exposure baselines. Sequences captured at 50% onboard-light intensity can additionally be expanded after capture using a co-located relighting model. To support reproducible acquisition, we introduce a software-controllable illuminator with programmable intensity control and a 60-degree beam angle. Together, CLPV , its lighting hardware, and its capture tooling provide a controlled testbed for studying how camera and illumination choices shape robot-vision measurements.

I. INTRODUCTION

Robot vision depends strongly on how images are acquired. In low-light and high-dynamic-range environments, underexposure, noise, blur, saturation, and reflections can degrade feature extraction and state estimation. Field-robotics datasets span diverse environments, trajectories, weather, and sensors [?], [?], [1]–[4], but typically treat illumination as an encountered condition rather than a controllable part of sensing.

The observed image depends on scene appearance and pose as well as exposure, gain, and active illumination. Exposure control can affect visual odometry and other robot-vision tasks [5]–[7]. Auxiliary light can improve feature tracking and visual estimation, while excessive light introduces saturation, reflections, and power cost [8]–[10]. Thus, “low light” does not define a single degraded condition: the same scene changes with the camera and illumination configuration.

Prior datasets vary exposure and gain in repeated observations [11], emulate exposure along robot trajectories [12], or repeat low-light traversals under auxiliary lighting [13]. These approaches demonstrate the value of controlled acquisition, but physical repeats scale poorly as sensing parameters multiply and cannot provide exact frame correspondence in viewpoint, motion, or scene state.

In this work, we present CLPV , a dataset designed to study controlled photometric variation for robot vision in low-

light environments. The dataset contains two complementary forms of data. First, we capture **136** fixed-view scenes under a controlled grid of **5 exposure times**, **4 gain settings**, and **11** co-located illumination intensities. This yields **220 hardware-synchronized stereo pairs** per scene while keeping scene geometry and viewpoint fixed. Each scene also includes an ambient lux reading and a measurement at every commanded light intensity from a co-located VEML7700 sensor. These observations allow the effects of exposure, gain, and active illumination to be analyzed independently of camera motion.

Second, CLPV contains **18** mobile-robot sequences comprising **13,667 hardware-synchronized stereo pairs**, together with LiDAR, odometry, and lux measurements spanning each complete traversal. We use a LiDAR-SLAM-based teach-and-repeat workflow to record the same route under complementary sensing baselines: fixed exposure with the co-located light at 50%, and camera auto-exposure along the repeated path. The fixed-exposure, 50%-light observations also serve as inputs to a co-located illumination decomposition model [8], which synthesizes alternative illumination levels while preserving the exact recorded viewpoint and motion.

Both components use a custom illumination platform with controllable intensity and ROS-based tools. Its electronics, firmware, and acquisition utilities will be released with the dataset.

We report the dataset composition, scene and depth coverage, and the observed camera and light response across the acquisition settings. The dataset supports controlled studies of how exposure and active illumination affect robot-vision measurements.

Our contributions are:

- 1) CLPV , currently ¹ comprising 136 fixed-view scenes over five exposures, four gains, and eleven light levels, plus 18 mobile sequences with synchronized stereo, lux, LiDAR, and odometry;
- 2) an illumination platform and associated capture software that provide remotely commanded light intensity and camera–light acquisition control;
- 3) post-capture illumination variants generated with CLID, together with a dataset-level characterization of photometric coverage, capture integrity, and route repeatability.

¹The counts in this workshop paper describe the current processed subset. Additional data is being collected; the intended release is approximately 500 fixed-view scenes and 50 robot sequences.

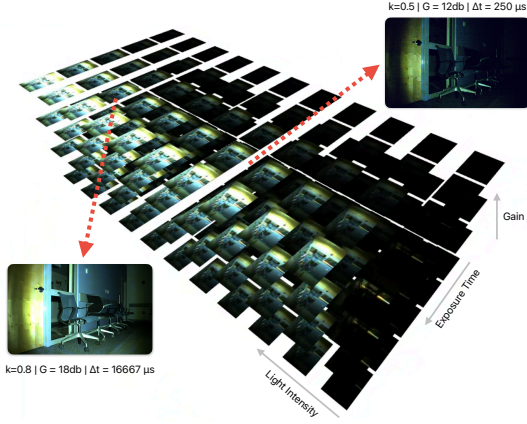


Fig. 2. Representative fixed-view sample from the controlled acquisition. The image tiles span $\mathcal{Q} = \Delta T \times \mathcal{G} \times \mathcal{K}$: exposure time, gain, and commanded co-located light intensity vary along the labeled axes, while scene geometry and camera pose remain fixed. Dashed callouts enlarge two cells at $k = 0.5$, $G = 12$ dB, $\Delta t = 250 \mu s$ and $k = 0.8$, $G = 18$ dB, $\Delta t = 16,667 \mu s$, illustrating the range within one scene sweep.

II. APPROACH

A. Acquisition Platform

We use one co-located camera–light assembly (Figure 1) for fixed-view and mobile acquisition. The light is mounted close to the camera axes and produces a 60-degree beam with approximately 3.4 kcd peak axial intensity from the robot’s 24-V supply. A co-located VEML7700 records illuminance. An Ouster OS-0 128-channel LiDAR provides point clouds for both components, while a ZED 2i stereo camera provides depth for fixed-view scenes using auto-exposure so that the photometric sweep does not affect depth estimation.

The controller combines a microcontroller, PWM dimmer, constant-current LED driver, electrical sensing, and closed-loop thermal protection. It accepts normalized intensity commands over USB serial, reports electrical and thermal state, and fails to a safe off state if host communication is lost. ROS, command-line, and graphical tools support manual control, power profiling, and scripted camera–light sweeps. The hardware, firmware, and host utilities will be released with the dataset.

Every observation is a hardware-synchronized stereo pair stored as one capture event. The release provides camera intrinsics, distortion, inter-camera extrinsics, and camera–light calibration following [?]. For mobile sequences, the LiDAR clock is synchronized to the acquisition Jetson using PTP; the timestamps are recorded with stereo, odometry, and sensing

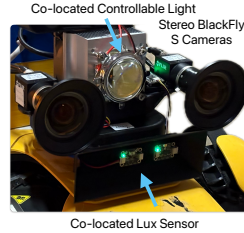


Fig. 1. Acquisition hardware with the controllable light, synchronized FLIR stereo cameras, and VEML7700 lux sensor.

state for cross-modal association. LiDAR-SLAM systems [?], [?] can therefore provide reference trajectories for evaluating visual SLAM across photometric settings.

B. Data Collection

a) *Controlled scenes*: The fixed-view acquisition varies camera exposure-time, gain, and co-located active illumination while holding scene geometry and viewpoint fixed. For each scene, the acquisition grid is

$$\begin{aligned} \Delta T &= \{\Delta t_1, \dots, \Delta t_5\}, \\ \mathcal{Q} &= \Delta T \times \mathcal{G} \times \mathcal{K}, \quad \mathcal{G} = \{g_1, \dots, g_4\}, \quad (1) \\ \mathcal{K} &= \{\ell_1, \dots, \ell_{11}\}, \end{aligned}$$

where ΔT , $\mathcal{G}$, and $\mathcal{K}$ denote five exposure times, four gain settings, and eleven commanded light intensities. The 220 configurations preserve scene geometry and camera pose. Each produces one synchronized stereo pair with camera readback and light state, distinguishing requested settings from realized values. Before each fixed-view sweep, we record ambient illuminance with the active light disabled. We then record a VEML7700 lux measurement for every commanded light intensity, providing an ambient reference and a measured light-response curve for each scene.

Figure 2 shows one sweep spanning severe underexposure, useful intermediate conditions, and saturation. The current subset contains 136 scenes; a capture is retained only when the scene remains static and its metadata confirms the expected camera and light state.

b) *Repeatable robot sequences*: The mobile subset adds realistic motion while reducing path variation. A teach pass records a reference route, which the robot follows during each repeat.

We record two complementary baselines on the repeated route. The reference run uses fixed camera exposure and a constant light command of $k_{\text{ref}} = 0.5$. This is the primary controlled sequence and is also the input condition used for CLID relighting. A second run enables the camera’s auto-exposure controller while retaining disabling the co-located light, providing the task-agnostic camera behavior commonly used in robot deployments. Hardware-synchronized stereo, PTP-timestamped LiDAR, odometry, camera state, light commands, and timestamps are recorded in both cases. VEML7700 lux measurements are recorded for the full duration of each sequence. The current subset contains 18 sequences and 13,667 stereo pairs; the release will additionally identify the taught route, repeat index, and synchronization status for every sequence.

C. Post-Capture Relighting

The fixed-exposure sequences are expanded along the illumination axis using the Co-Located Illumination Decomposition (CLID) model from [8]. CLID predicts an ambient image A , a scalar co-located light-contribution map S_F , and a global light-color vector C_F . Following its original formulation, an image at target light intensity k is synthesized as

$$I^k = A + k(S_F \oplus C_F), \quad k \in \mathcal{K} \subseteq [0, 1], \quad (2)$$

where $\oplus$ applies the global color vector to the scalar light map. The operation is defined in linear sensor space under the fixed capture exposure. For each frame recorded at $k_{\text{ref}} = 0.5$, we generate the requested discrete levels in $\mathcal{K}$. These variants preserve the exact camera motion, frame timing, scene state, LiDAR, and odometry of the source sequence; only the modeled active-light contribution changes. The auto-exposure repeat captures a real camera controller on a closely repeated route, whereas CLID isolates illumination changes on the same frames.

We plan to extend CLID to synthesize observations at different camera exposures by scaling the reconstructed image using a normalized exposure ratio. Let e_{ref} denote the effective exposure of the input image and e the target exposure, where effective exposure combines shutter time and linear sensor gain. Assuming a locally linear camera response, the extended formulation is

$$I^{k,e} = \text{clip}_{[0, I_{\text{max}}]}(\rho_e [A + k(S_F \oplus C_F)]), \quad (3)$$

$$\rho_e = \frac{e}{e_{\text{ref}}}, \quad k \in \mathcal{K} \subseteq [0, 1].$$

Here, e/e_{ref} is the normalized exposure ratio and I_{max} the sensor saturation value. The physical sweeps in CLPV provide corresponding observations for validating this first-order approximation.

III. DATASET CHARACTERIZATION

CLPV is a controlled data resource rather than a benchmark tied to one algorithm. We characterize its sensing coverage, capture integrity, and route repeatability while leaving task-specific evaluation to dataset users.

A. Composition and Coverage

Table I summarizes the processed subset. The release manifest will include requested and reported camera settings, light level, VEML7700 measurements, stereo trigger identifier, timestamps, calibration, checksums, and a validity flag for every expected stereo pair.

Figure 3 characterizes the fixed-view collection. For each scene with valid Z-camera depth, we compute the median and 95th-percentile range. GPT-5.6 Luna classifies the corresponding images by indoor environment and assigns ordered texture and reflectance levels. These generated labels are dataset descriptors, not task ground truth. The current set spans offices, laboratories, corridors, atria, residential, transitional, and utility spaces, including compact scenes and long-range corridor or atrium views. Mobile sequences will additionally be summarized by environment, route, repeat count, duration, distance, and motion after the dataset is finalized.

TABLE I
CURRENT COMPOSITION OF CLPV. COUNTS WILL BE UPDATED FOR THE FROZEN RELEASE.

Quantity	Current value
Fixed-view scenes	136
Exposure settings	5
Gain settings	4
Physical light settings	11 (0–100%)
Settings per fixed scene	220
Expected fixed stereo pairs	29,920
Expected fixed images	59,840
Mobile sequences	18
Mobile stereo pairs	13,667
Physical sequence baselines	Fixed exposure; auto exposure
Reference light level	50%
Fixed-view lux measurements	Ambient and per light intensity
Auxiliary mobile streams	LiDAR, odometry, and VEML7700 lux
Synchronization	Stereo hardware trigger; LiDAR–Jetson PTP
Calibration	Full stereo intrinsics, distortion, and extrinsics

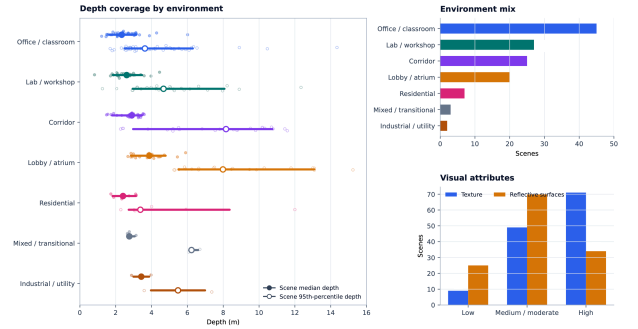


Fig. 3. Coverage of fixed-view scenes with valid depth and classification outputs. Left: per-scene median and 95th-percentile depth by GPT-5.6 Luna environment class. Right: environment counts and VLM-assigned texture and reflectance levels. The semantic attributes are dataset descriptors, not evaluation ground truth.

B. Photometric Range

Figure 4 summarizes image statistics across both FLIR cameras. Exposure is expressed in stops relative to 1/60 s at 0 dB, combining shutter time and gain. As exposure and commanded light increase, observations move from near-black toward higher mean intensity, stronger gradients, and eventually saturation. The light has the clearest visible effect at intermediate exposures; the lowest settings remain signal-limited and the highest increasingly clip. The plots therefore confirm a measurable camera response to the co-located light and summarize the useful photometric range of the selected scenes.

C. Capture Integrity

We audit observations, sequences, and sensor streams. Fixed-view checks cover grid and stereo-pair completeness, trigger consistency, setting readback, light settling, corrupted

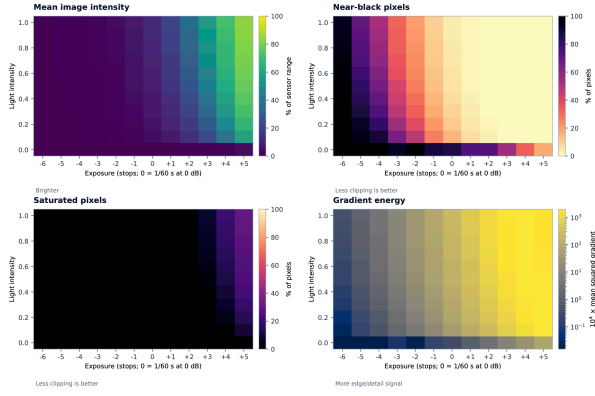


Fig. 4. Image response across the fixed-view acquisition grid and both FLIR cameras. The panels report mean intensity, near-black pixels, saturated pixels, and gradient energy over exposure and commanded light intensity.

files, and scene motion. Mobile checks cover stereo continuity, dropped frames, LiDAR and odometry rates, PTP offset, cross-modal timestamp residuals, and valid measurement association. Checksums and the audit script will be released with the manifest.

All left–right pairs use the shared hardware trigger; mobile LiDAR is synchronized to the acquisition Jetson using PTP. We report measured timing residuals and PTP offset, and do not treat robot odometry as ground truth.

D. Route Repeatability and Sequence Baselines

Using LiDAR-SLAM localization, we align each repeat to its taught route and report cross-track displacement, heading difference, path-length ratio, and speed distribution. These geometric statistics measure how closely the fixed- and auto-exposure runs revisit the same support without treating physical traversals as frame-identical.

For each route, the release groups the fixed-exposure 50%-light run, its auto-exposure repeat, and CLID-derived variants. A sequence card will show the LiDAR-map route, representative frames, camera-state traces, and relit examples without prescribing a downstream method or metric.

IV. DISCUSSION

The two acquisition modes provide complementary forms of control. A fixed-view sweep isolates photometric settings precisely but does not contain robot motion. Physical route repeats include realistic motion and the behavior of an actual auto-exposure controller, but they cannot reproduce camera pose or scene timing exactly. CLID variants provide exact frame correspondence across light levels, but they are derived data and should not be presented as independent physical traversals. The release keeps these provenance classes explicit so that users can select the form of control appropriate to their study.

The controlled grids support analysis of feature stability, enhancement, exposure selection, and illumination-aware image

formation while holding geometry fixed. Repeated physical sequences support fixed- versus auto-exposure comparisons under closely matched motion, while CLID variants provide multiple light levels on the same frames for tracking, mapping, or learned-perception studies. These are intended uses rather than a fixed leaderboard.

Several measurements remain important before the dataset is frozen. The commanded light level is a normalized controller input rather than a physical output measurement. The co-located VEML7700 records local illuminance in lux, but it does not measure irradiance at every scene surface. The release will therefore provide both the commanded level and measured lux: an ambient reading and per-intensity measurements for fixed-view scenes, and a lux trace spanning each mobile sequence. We will also report camera–light calibration, hardware-trigger and PTP timing residuals, stereo-pair completeness, and route-repeatability statistics. The released stereo calibration provides the intrinsics, distortion, and inter-camera transform required for reconstruction. Counts in this draft describe the processed subset; collection is ongoing toward approximately 500 fixed-view scenes and 50 robot sequences, and final counts will be regenerated from the release manifest.

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